

Development Roadmap - RAG APP

A production-level RAG application using:

- **Frontend:** React + Vite
- **Backend:** Node.js + Express
- **Vector DB:** ChromaDB , LangChain
- **LLM:** Gemini/OpenAI
- **Database:** MongoDB
- **Deployment:** Docker

Phase 1: Project Setup

- ☒ Create React + Vite frontend
- ☒ Create Node + Express backend
- Configure Tailwind CSS
- Install required dependencies
- Configure environment variables
- Connect frontend and backend
- Verify the application runs

Phase 2: Backend Foundation

- MongoDB connection
- Express server
- Global error handling
- Logging
- API structure
- File upload (Multer)

Phase 3: Document Processing

- Upload PDF/DOCX/TXT
- Extract text
- Chunk the text
- Generate embeddings
- Store embeddings in ChromaDB

Phase 4: RAG Pipeline

- User asks a question
- Convert question to embedding

- Retrieve relevant chunks
- Create prompt
- Send to Gemini
- Return answer with citations

Phase 5: Frontend

- Beautiful landing page
- Upload documents
- Chat interface
- Chat history
- Loading states
- Source references
- Error handling

Phase 6: Authentication (Optional)

- JWT Authentication
- Login
- Signup
- User-specific documents

Phase 7: Deployment

- Dockerize frontend
 - Dockerize backend
 - Docker Compose
 - Deploy frontend
 - Deploy backend
 - Configure production environment
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How we'll build it

I suggest we build it **feature by feature**, not file by file.

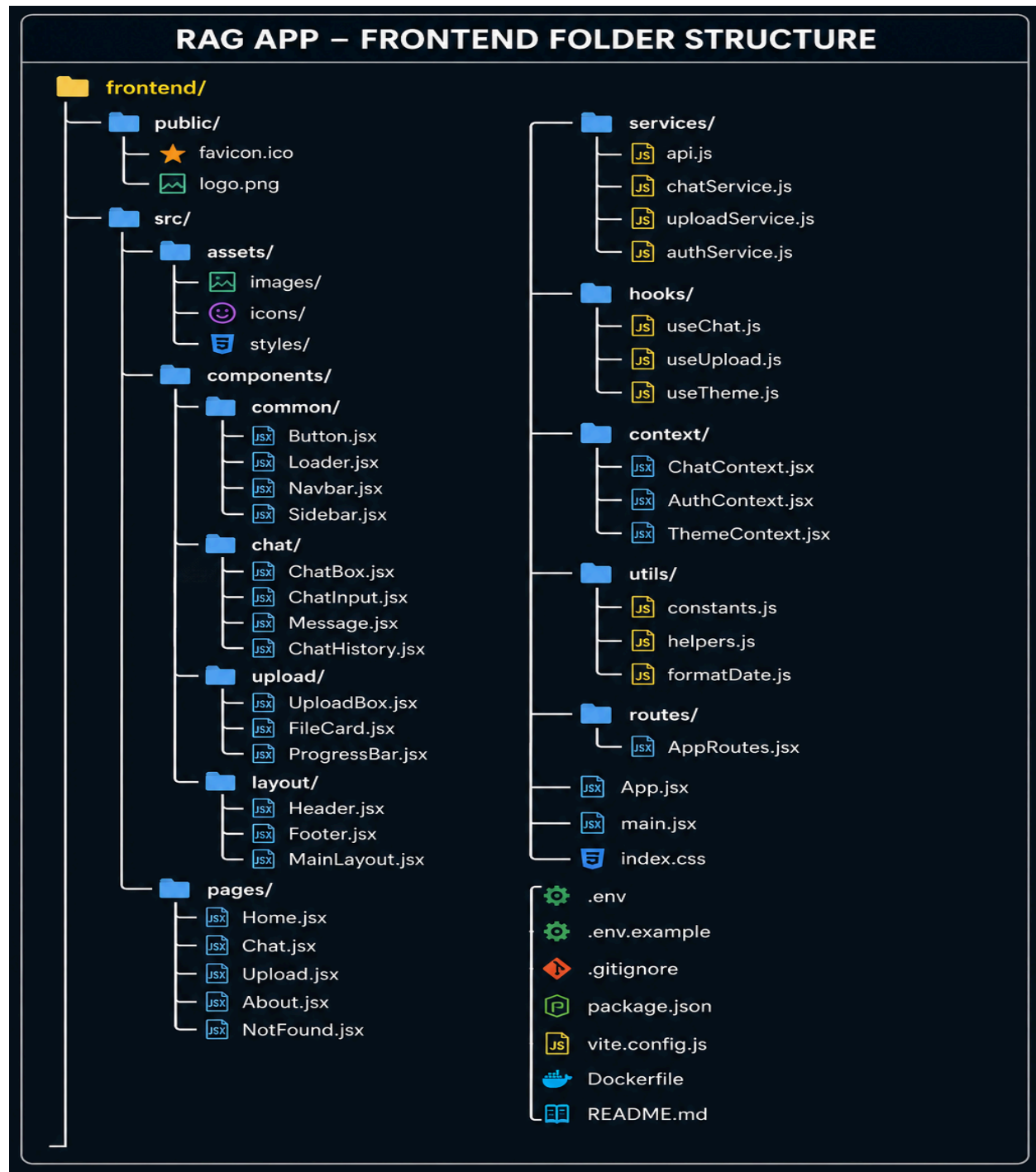
For every feature, we'll follow this sequence:

1. Understand the concept.
2. Create the required folders/files.
3. Write the code.
4. Test it.
5. Explain why it works.
6. Commit it to Git.

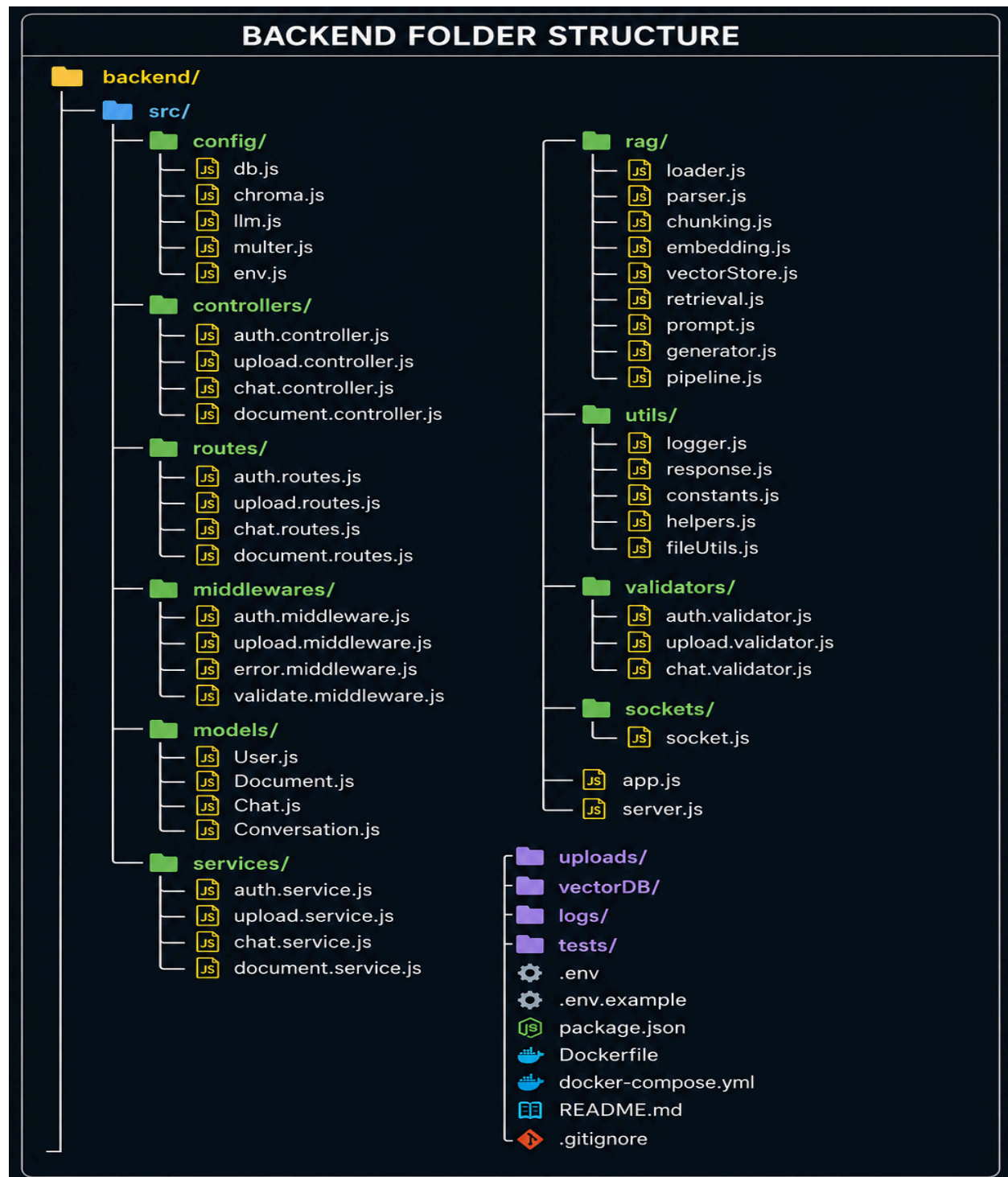
Production-Level Structure



React + Vite Folder Structure



Production-Level Backend Structure



Suggested Packages - FRONTEND

npm install react-router-dom
npm install axios
npm install react-hot-toast
npm install react-icons
npm install react-markdown
npm install remark-gfm
npm install lucide-react
npm install framer-motion (optional - for animations)

Suggested Packages - BACKEND

Recommended Packages :

npm install express
npm install mongoose
npm install dotenv
npm install cors
npm install multer
npm install axios
npm install jsonwebtoken
npm install bcryptjs
npm install cookie-parser
npm install express-validator
npm install uuid
npm install morgan

RAG Packages

npm install langchain
npm install @langchain/community
npm install chromadb
npm install pdf-parse
npm install mammoth
npm install dotenv

Gemini

npm install @google/genai